

Organizing Panel Data Methods

From Homogeneous Slopes to Cross-Sectional Dependence and Structural Breaks

A Map of the Whole Field — for Beginners

The purpose of these notes

Most courses teach panel methods as a *list*. This is a **map**. Every method here exists because it **relaxed one assumption** of the method before it. Once you see the chain — Pooled → FE → MG → PMG → CCE → breaks → the frontier — you can place any new paper on it, and you can *choose* rather than guess.

Each method gets the same four questions: **What is the idea?** **When do I use it?** **When must I not?** **How does it differ from its neighbour?**

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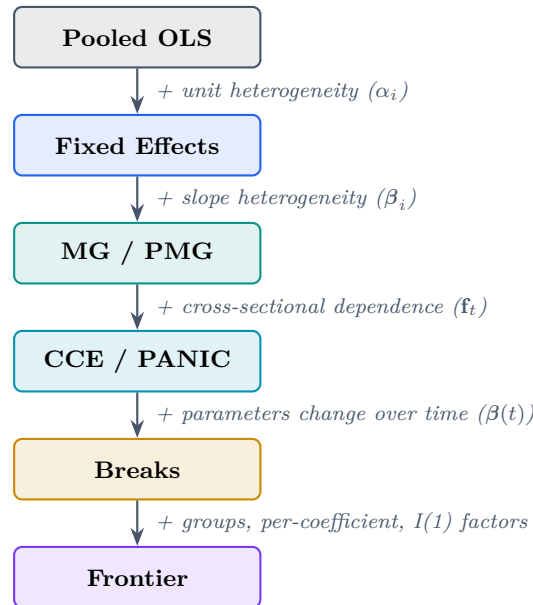
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1 The map: how every method connects

Read this page first

There is **one** idea behind the whole field. Start with the most restrictive model imaginable, then relax **one assumption at a time**. Each relaxation buys realism and costs parameters. Every method in this document is one step on that chain.



1.1 The five questions that decide everything

#	Question	If the answer is “yes” you must move to
1	Do units have their own intercepts ?	Fixed Effects (Section 2)
2	Do units have their own slopes ?	MG / PMG (Section 3)
3	Do units share common shocks ?	CCE / PANIC (Section 5)
4	Are the series non-stationary ?	unit-root & cointegration tests (Section 6)
5	Do the parameters change over time ?	structural-break methods (Sections 7–8)

For beginners

Notice these are *cumulative*, not alternatives. A frontier method like `xtkpybreak` answers **all five** at once: unit effects, heterogeneous slopes, common (even $I(1)$) factors, non-stationarity, and breaks. That is why it looks complicated — it is five ideas stacked.

2 The homogeneous world

2.1 Pooled OLS — the starting point

Pooled OLS

$$y_{it} = \alpha + \mathbf{x}'_{it}\beta + \varepsilon_{it}$$

The idea

Ignore that the data is a panel. Treat all NT observations as one big cross-section: everyone shares the same intercept α and the same slope β .

When to use it

Only when there is genuinely **no unobserved unit-specific heterogeneity** — or when it is uncorrelated with $\mathbf{x}$ *and* you do not care about efficiency. In practice: as a **benchmark** to compare against, almost never as the final model.

When NOT to use it

Whenever α_i exists and correlates with $\mathbf{x}_{it}$ — which is the normal case. Then pooled OLS suffers **omitted-variable bias**: the estimated slope mixes the within-unit relationship with the between-unit one, and can even have the *wrong sign*.

2.2 Fixed Effects — letting intercepts differ**Fixed Effects: the within transformation**

$$y_{it} = \alpha_i + \mathbf{x}'_{it}\boldsymbol{\beta} + \varepsilon_{it} \implies \ddot{y}_{it} = \ddot{\mathbf{x}}'_{it}\boldsymbol{\beta} + \ddot{\varepsilon}_{it}, \quad \ddot{y}_{it} = y_{it} - \bar{y}_i.$$

The idea

Subtract each unit's own time-mean. Anything constant within a unit — including α_i — vanishes. What is left is the **within-unit** relationship.

When to use it

When you believe α_i is correlated with $\mathbf{x}_{it}$ (the usual case) and your question is about **within-unit** variation: “when a country's x rises, what happens to *its* y ?”

When NOT to use it

When you need coefficients on **time-invariant** regressors — FE wipes them out. And in **dynamic** models with small T : the within transformation correlates $y_{i,t-1}$ with the demeaned error, giving **Nickell bias** of order $1/T$.

2.3 Random Effects and the Hausman test**Random Effects**

$$y_{it} = \alpha + \mathbf{x}'_{it}\boldsymbol{\beta} + \underbrace{u_i + \varepsilon_{it}}_{\text{composite error}}, \quad \text{Cov}(u_i, \mathbf{x}_{it}) = 0$$

RE treats u_i as *random* and estimates by GLS (quasi-demeaning with $\theta = 1 - \sqrt{\sigma_\varepsilon^2 / (T\sigma_u^2 + \sigma_\varepsilon^2)}$).

FE vs RE — the only real question

Both estimate the same $\boldsymbol{\beta}$. The difference is a single assumption: **is α_i correlated with $\mathbf{x}_{it}$?**

- **Not correlated** $\Rightarrow$ RE is *consistent and more efficient*, and can estimate time-invariant regressors.
- **Correlated** $\Rightarrow$ RE is *inconsistent*; FE is still consistent.

The **Hausman test** compares them:

$$H = (\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{RE})' [V(\hat{\boldsymbol{\beta}}_{FE}) - V(\hat{\boldsymbol{\beta}}_{RE})]^{-1} (\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{RE}) \xrightarrow{d} \chi_K^2$$

H_0 : RE is consistent. **Reject $\Rightarrow$ use FE.**

The modern caveat on Hausman

The classic Hausman test assumes **homogeneous slopes** and **no cross-sectional dependence**. In macro panels both usually fail — which is exactly why the rest of this document exists. Do not let a Hausman test be the end of your specification search.

2.4 Comparing the homogeneous family

Estimator	Assumes about α_i	Slope	Time-inv. x ?	Main risk
Pooled OLS	none ($\alpha_i = \alpha$)	common	yes	omitted heterogeneity
Fixed Effects	parameter, may correlate	common	no	loses time-invariant vars
Random Effects	random, uncorrelated	common	yes	inconsistent if assumption fails
Between	(uses unit means)	common	yes	discards within variation

3 Slope heterogeneity: MG, PMG, DFE**The assumption we now drop**

So far β was **the same for every unit**. In a panel of countries that is a strong claim: does a 1% rise in energy use really raise GDP by the same amount in Norway and in Chad? Now we let β_i differ:

$$y_{it} = \alpha_i + \mathbf{x}'_{it}\beta_i + \varepsilon_{it}$$

3.1 Mean Group (MG) — Pesaran & Smith (1995)**Mean Group**

$$\hat{\beta}_{MG} = \frac{1}{N} \sum_{i=1}^N \hat{\beta}_i, \quad \hat{V}(\hat{\beta}_{MG}) = \frac{1}{N(N-1)} \sum_{i=1}^N (\hat{\beta}_i - \hat{\beta}_{MG})(\hat{\beta}_i - \hat{\beta}_{MG})'$$

The idea

Run a **separate regression for each unit**, then simply **average** the slopes. The variance comes from how much the unit slopes *disagree*.

When to use it

When slopes are genuinely heterogeneous and T is **large enough** to estimate each unit's regression on its own (rule of thumb $T \gtrsim 20-30$). It is fully flexible: it assumes nothing about how slopes relate.

When NOT to use it

When T is small — each $\hat{\beta}_i$ is then noise, and averaging noise gives an imprecise (though still unbiased) mean. Also sensitive to **outlier units**: one crazy country can move the average.

3.2 Pooled Mean Group (PMG) — Pesaran, Shin & Smith (1999)**PMG: an error-correction model with a pooled long run**

$$\Delta y_{it} = \phi_i(y_{i,t-1} - \theta' \mathbf{x}_{i,t-1}) + \sum_j \lambda_{ij} \Delta y_{i,t-j} + \sum_j \delta'_{ij} \Delta \mathbf{x}_{i,t-j} + \varepsilon_{it}$$

The idea

A **compromise**. Notice θ has *no* i subscript but ϕ_i and the short-run terms do: the **long-run relationship is forced to be the same** for all units, while the **speed of adjustment and short-run dynamics are free**.

Why this compromise is so popular

Economic theory usually predicts a **common long run** (a shared equilibrium: PPP, a production function, an energy–growth elasticity) but says nothing about how fast each country gets there. PMG encodes exactly that belief — and by pooling the long run it gains a lot of efficiency over MG.

When to use it

When theory suggests a **common long-run relationship** but heterogeneous adjustment — and the poolability restriction is **not rejected** by a Hausman test.

When NOT to use it

When the long-run homogeneity restriction is **false**: PMG is then inconsistent. Always test it (below). Also note $\phi_i < 0$ is required for a meaningful error-correction interpretation.

3.3 Dynamic Fixed Effects (DFE)**The idea**

Pool **everything** except the intercepts: long run, short run, and adjustment speed are all common. The most restrictive dynamic option.

When NOT to use it

Badly biased when slopes truly differ (Pesaran–Smith show heterogeneity bias does *not* disappear as $T \rightarrow \infty$ in dynamic models). Use it as a **benchmark**, rarely as the answer.

3.4 The comparison you must know: MG vs PMG vs DFE

What is pooled?	MG	PMG	DFE
Intercepts α_i	free	free	free
Long-run θ	free	pooled	pooled
Adjustment speed ϕ_i	free	free	pooled
Short-run dynamics	free	free	pooled
Consistent if slopes differ?	yes	only if LR is truly common	no
Efficiency	lowest	middle	highest (if true)
Needs large T ?	yes (per unit)	moderate	moderate
Stata	xtmg	xtpmg, pmg	xtpmg, dfe

How to choose between MG and PMG — the Hausman poolability test

Under long-run homogeneity **both** MG and PMG are consistent but PMG is *efficient*; under heterogeneity **only** MG is consistent. That is exactly the Hausman setup:

$$H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})' [V(\hat{\theta}_{MG}) - V(\hat{\theta}_{PMG})]^{-1} (\hat{\theta}_{MG} - \hat{\theta}_{PMG}) \xrightarrow{d} \chi^2$$

H_0 : the long run is poolable. **Do not reject** $\Rightarrow$ report **PMG** (more efficient). **Reject** $\Rightarrow$ report **MG**.

```

xtset id year
xtpmg d.y d.x, lr(1.y 1.x) ec(ec) replace pmg // pooled long run
estimates store pmg
xtpmg d.y d.x, lr(1.y 1.x) ec(ec) replace mg // fully heterogeneous
estimates store mg
hausman mg pmg, sigmamore // H0: long run is poolable

```

4 A note on dynamics: why FE breaks down

The dynamic panel and Nickell bias

$$y_{it} = \rho y_{i,t-1} + \mathbf{x}'_{it}\beta + \alpha_i + \varepsilon_{it}, \quad \text{Bias}(\hat{\rho}_{FE}) = O\left(\frac{1}{T}\right)$$

The idea

The within transformation subtracts $\bar{y}_i$, which *contains* ε_{it} . So the demeaned regressor $\tilde{y}_{i,t-1}$ is correlated with the demeaned error by construction. The bias is $O(1/T)$ — it vanishes only as $T \rightarrow \infty$.

Situation	Use	Stata
Small T , large N	GMM : Arellano–Bond (difference), Blundell–Bond (system)	<code>xtabond</code> , <code>xtdpdsys</code>
Large T	MG / PMG (bias is small)	<code>xtpmg</code> , <code>xtmg</code>
Large T + dependence	dynamic CCE / CS-ARDL	<code>xtdcce2</code>
Dependence + breaks	robust dynamic estimators	<code>xtdynestimb</code>

5 Cross-sectional dependence: the great divide

The assumption we now drop

Every method so far assumed units are **independent**. They are not: oil shocks, crises and technology hit everyone. Model it with a **common factor** error:

$$y_{it} = \mathbf{x}'_{it}\beta_i + u_{it}, \quad u_{it} = \gamma'_i \mathbf{f}_t + \varepsilon_{it}$$

$\mathbf{f}_t$ = unobserved global shocks; γ_i = how exposed unit i is.

What happens if you ignore it

Pooled/FE/MG standard errors are wrong and tests **over-reject**: with a shared factor there is far *less* independent information than N , but the formulas divide by N anyway. You will “find” relationships that are artefacts of the common shock.

5.1 Step 1: test for it

Pesaran CD test

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i < j} \hat{\rho}_{ij} \xrightarrow{d} N(0, 1)$$

Test	Valid when	Weakness
Breusch–Pagan LM	small N , large T	explodes for large N
Scaled LM	N, T large	size distortion if $N \gg T$
Bias-corrected LM	large N, T (FE)	needs the FE structure
Pesaran CD	large N , any T	signed average can cancel out
CD*	large N (de-factored residuals)	needs $\hat{\tau}$
Frees / Friedman	small–moderate N	non-parametric, lower power
α exponent	measures <i>strength</i> , not a test	–

Classic vs modern question

Classic: “is there dependence — yes or no?” *Modern*: “**how strong** is it?”, indexed by α in $\sum_i |\gamma_i| = O(N^\alpha)$: $\alpha = 1$ **strong** (a real factor $\Rightarrow$ CCE/PANIC); $\frac{1}{2} < \alpha < 1$ **semi-weak**; $\alpha \leq \frac{1}{2}$ **weak/spatial** (robust s.e. may suffice).

5.2 Step 2: remove it — the two families

5.2.1 Family A: cross-sectional averages (CCE)

CCE — Pesaran (2006)

$$\hat{\beta}_i = (X_i' \bar{M} X_i)^{-1} X_i' \bar{M} y_i, \quad \bar{M} = I - \bar{Z}(\bar{Z}' \bar{Z})^{-1} \bar{Z}', \quad \bar{Z} = [\bar{y}, \bar{X}]$$

$$\hat{\beta}_{CCEMG} = \frac{1}{N} \sum_i \hat{\beta}_i \quad (\text{Mean Group}); \quad \hat{\beta}_{CCEP} \text{ (Pooled)}$$

The idea

Do not estimate the factors. Their footprint is already in the cross-section averages $\bar{y}_t, \bar{\mathbf{x}}_t$ — so add those as regressors and project them out. You never need to know *how many* factors there are.

When to use it

The default in modern macro panels. Especially when you want **heterogeneous slopes**, do not want to commit to a number of factors, and — crucially — when the factors may be **I(1)** (Kapetanios–Pesaran–Yamagata 2011 prove CCE survives this).

When NOT to use it

When N is small (the averages are noisy), or the **rank condition** fails ($\text{rank}(\bar{\Gamma}) = r \leq k + 1$: you need at least as many observables as factors).

The CCE family tree.

Member	What it adds	Reference
CCEMG	averages the unit slopes; heterogeneous	Pesaran (2006)
CCEP	pools the slopes; efficient if homogeneous	Pesaran (2006)
Dynamic CCE	extra lags of the averages (needed if dynamic)	Chudik–Pesaran (2015)
CS-ARDL	dynamics + averages; long run <i>and</i> short run	Chudik et al.
CS-DL	estimates the long run <i>directly</i> , robust to lag misspecification	Chudik et al.
F-CCEMG	+ unit-specific Fourier terms $\Rightarrow$ absorbs heterogeneously-timed breaks	Guliyev (2026)
CCE + break	+ a common structural break (and I(1) factors)	BFK; KPY; BFW

5.2.2 Family B: principal components (PANIC / IFE)

PANIC — Bai & Ng (2004)

$$\Delta y_{it} = \gamma'_i \Delta \mathbf{f}_t + \Delta e_{it} \xrightarrow{\text{PCA on differences}} \hat{\mathbf{f}}_t = \sum_{s \leq t} \Delta \hat{\mathbf{f}}_s, \quad \hat{e}_{it} = \sum_{s \leq t} \Delta \hat{e}_{is}$$

The idea

Estimate the factors explicitly. You cannot run PCA on levels when $\mathbf{f}_t$ and e_{it} have different integration orders — so difference first, extract, then re-cumulate, and test the common and idiosyncratic parts *separately*. Choose r by Bai–Ng: $IC_{p2}(k) = \ln V(k, \hat{F}^k) + k \frac{N+T}{NT} \ln \min\{N, T\}$.

CSA vs factors — the comparison to remember

Aspect	CCE (averages)	PANIC / IFE (factors)
Estimates the factors?	no — only proxies them	yes — by PCA
Need to know r?	no (big practical win)	yes (Bai–Ng IC)
Key assumption	rank condition	enough factors; large N, T
Works with I(1) factors?	yes (KPY 2011)	yes (differencing)
Slope heterogeneity	natural (CCEMG)	usually pooled
What it tells you	a clean slope	plus whether non-stationarity is common or idiosyncratic
Typical commands	<code>xtdcce2</code> , <code>xtkpybreak cce</code> , <code>xtccoint</code>	<code>xtpcointegwe</code> , <code>xtbreakcoint</code>

They are cousins, not rivals: many third-generation commands use *both*.

6 Non-stationarity: the generations

The idea

If the series are I(1), a levels regression can be **spurious**. So we test: is there a unit root? and do the series **cointegrate**? The tests are sorted into three **generations** by which assumption each finally drops.

Characteristic	1st gen	2nd gen	3rd gen
Cross-sectional dependence	no	yes	yes
Structural breaks	no	no	yes (estimat
How dependence is handled	—	CCE / factors / bootstrap	same + break
Critical values	standard	factor/bootstrap	depend on model and
Failure if misused	over-rejects under CSD	loses power under breaks	(heavier to com
Unit-root example	IPS, LLC	CIPS, PANIC	Carrion et al.; xt
Cointegration example	Pedroni, Kao	Westerlund (2007)	Banerjee–Carrion; xt

The null-direction trap — the most common error

Half these tests have $H_0 = \text{cointegration}$ (`xtlmbreak`, `xtpcointegboot`) and half $H_0 = \text{no cointegration}$ (`xtpcointegwe`, `xtbreakcoint`, `xtcceoint`, `xtcadfcoint`). “Reject” means **opposite** things. State the null before the p-value.

Confirmatory analysis

Unit-root tests have low power. Best practice: run **both** an ADF-type (H_0 : unit root) and a KPSS-type (H_0 : stationarity). If they agree, the conclusion is robust; if both reject or neither does, say the data are uninformative.

7 Structural breaks: organizing the ideas

The assumption we now drop

Everything so far assumed the parameters are **constant over time**. Now let β depend on t :

$$\beta(t) = \beta_j \quad \text{for } T_{j-1} < t \leq T_j$$

7.1 Axis 1 — who breaks? (across units)

Type	Restriction	Schedules	Method
Common	$T_{B1} = \dots = T_{BN}$	1	Qian–Su; BFK; <code>xtkpybreak</code>
Grouped (latent)	$T_{Bi} = T_g$ if $i \in G_g$	G	GAGFL <code>xtbreakmodel</code> , <code>gagfl</code>
Heterogeneous	$T_{Bi} \neq T_{Bj}$	N	F-CCEMG <code>xtfmg</code> <code>fccemg</code>

The difference

Fewer schedules $\Rightarrow$ fewer parameters $\Rightarrow$ **more power if true**, but **inconsistency if false**. Common $\subset$ Grouped $\subset$ Heterogeneous. Grouped is often the sweet spot: it *nests* the common case ($G = 1$), so allowing it costs you nothing.

7.2 Axis 2 — what breaks? (inside the model)

Break in	Meaning	Command
intercept α_i	the level shifts	<code>xtpkpss</code>
slope $\beta(t)$	the marginal effect changes	<code>xtbreakmodel</code> , <code>xtcbc</code>
cointegrating vector	the long-run equilibrium moves	<code>xtbreakcoint</code> , <code>xtcadfcoint</code>
factor loadings $\gamma_i(t)$	sensitivity to the global shock changes	<code>xtkpybreak</code>

7.3 Axis 3 — which coefficients? (vector vs CBC)

The difference

Vector break: all K coefficients jump together at each date (Bai–Perron, Qian–Su).

Coefficient-by-coefficient: each β_k has its *own* m_k breaks and dates (Kaddoura 2025).

Why it matters: a vector break splits *every* coefficient at the union of dates, even ones that never moved — inflating variance and hiding significance. In Kaddoura’s crime application, vector-break finds 2 breaks in **all 16** coefficients; CBC finds them in **only 3**.

7.4 Axis 4 — how is it modelled? (the three families)

Family	How	Best when	Commands
Dummy	step at an estimated date	a clear policy date / crisis	<code>xtpkpss</code> , <code>xtlmbreak</code>
Fourier	sine/cosine terms, no date	gradual / unknown number	<code>xtpqroot</code> , <code>fourier</code> , <code>xtpfardl</code>
Regularization	penalise change; keep survivors	many candidates, automatic selection	<code>xtbreakmodel</code> , <code>xtcbc</code>

7.4.1 Regularization in one page (the modern school)

Adaptive group fused lasso

$$\min_{\{\beta_t\}} \frac{1}{N} \sum_{i,t} (y_{it} - \mathbf{x}'_{it} \beta_t)^2 + \lambda \sum_{t=2}^T w_t \|\beta_t - \beta_{t-1}\|, \quad w_t = \|\dot{\beta}_t - \dot{\beta}_{t-1}\|^{-\kappa}$$

The idea

Let β change **every** period, then **penalise the differences**. Most differences are shrunk to *exactly* zero (no break there); the survivors **are** the break dates. So break detection becomes automatic variable selection — you never specify how many breaks.

The two refinements

Group: penalise the vector norm $\Rightarrow$ all K break together. **Adaptive:** weight by the inverse of a pilot estimate $\Rightarrow$ real jumps are barely penalised, noise is heavily penalised. This delivers the **oracle property**: the method asymptotically finds the true breaks *as if the dates were known*. λ is chosen by a BIC-type IC — too small over-segments, too large over-smooths.

8 The frontier: putting it all together

Method	What it combines	Command
GAGFL	heterogeneous slopes + latent groups + fused lasso breaks	<code>xtbreakmodel, gagfl</code>
CBC	fixed- T + per-coefficient breaks + interactive effects	<code>xtcbc</code>
KPY + BFW	CCE + I(1) factors + breaks in slopes <i>and</i> loadings	<code>xtkpybreak</code>
BFK	CCE + common break + endogenous regressors	<code>xtbfkbreak</code>
F-CCEMG	CCE + Fourier $\Rightarrow$ heterogeneously-timed breaks	<code>xtfmg fccemg</code>
Fourier CS-ARDL	ARDL long/short run + Fourier + cross-section averages	<code>xtpfardl</code>
Shrinkage QR	quantile regression + fused lasso breaks	<code>xtquantilebreak</code>
Banerjee–Carrion 2025	cointegration + breaks in deterministics, vector and loadings	<code>xtcadfcoint</code>

The frontier is not new ideas — it is combinations

Look at the middle column. Every frontier method is just **two or three of the five relaxations stacked**. `xtkpybreak` = heterogeneous slopes + CCE + I(1) factors + breaks. `xtcbc` = fixed- T + breaks + interactive effects. Once you know the five axes, no paper is mysterious — you simply ask *which axes did they relax?*

9 The decision map

1. Is α_i correlated with $\mathbf{x}$? $\rightarrow$ FE (Hausman).
2. Are slopes heterogeneous? $\rightarrow$ MG. Is the *long run* poolable (Hausman)? $\rightarrow$ PMG.
3. Is there CSD? (`xtcd2`) $\rightarrow$ if yes, CCE / PANIC. Dynamic? $\rightarrow$ CS-ARDL.
4. Are the series I(1)? $\rightarrow$ unit-root tests *with* breaks and CSD (`xtpkpss`, `xtpqroot`).
5. Do they cointegrate? $\rightarrow$ break-robust cointegration tests (`xtpcointegwe`, `xtbreakcoint`, `xtcadfcoint`).
6. Do parameters break? $\rightarrow$ which axis?
 - same date for all $\rightarrow$ `xtbreakmodel, pls` / `xtkpybreak`
 - hidden groups $\rightarrow$ `xtbreakmodel, gagfl`
 - every unit different $\rightarrow$ `xtfmg fccemg`
 - only some coefficients $\rightarrow$ `xtcbc`
 - gradual $\rightarrow$ `xtpqroot`, `fourier` / `xtpfardl`
 - loadings too $\rightarrow$ `xtkpybreak break, loadings`
7. **Interpret:** match every estimated date to a real event, with a confidence interval.

```
* ---- the organized workflow, in order ----
webuse grunfeld, clear
xtset company year

xtcd2 invest // 3. is there CSD?
xtpkpss invest, model(constbreak) maxbreaks(2) // 4. unit root WITH breaks
xtpcointegwe invest mvalue, model(regimeshift) maxfactors(2) // 5. cointegration WITH breaks
```

```
xtfm map invest mvalue kstock // which estimator does my panel need?
xtbreakmodel invest mvalue, method(pls) // 6. common breaks
xtcbc invest mvalue kstock, graph // only some coefficients?
xtkpybreak break invest mvalue kstock, nbreaks(1) // CCE + I(1) factors + breaks
```

The one sentence

Panel econometrics is a **ladder of relaxed assumptions**: common intercepts → own intercepts → own slopes → shared shocks → non-stationarity → parameters that change. Every estimator is a rung. **Climb only as high as your data forces you** — each rung costs parameters and power.

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